

STAT230A Lecture 15 Coefficient of Determination

1 R^2 : Multiple Correlation Coefficient

1.1 Definition: Pearson's Correlation

令 $x_i, y_i \in \mathbb{R}$, $i = 1, \dots, n$, 则 Pearson's correlation 被定义为:

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^n (x_i - \bar{x})^2}} \in [-1, 1]$$

1.2 Definition: R^2 (Scalar Version)

令 $x_i, y_i \in \mathbb{R}$, $i = 1, \dots, n$, 则 R^2 between x_i and y_i 被定义为:

$$R^2 = \hat{\rho}_{xy}^2 = \frac{(\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x}))^2}{(\sum_{i=1}^n (y_i - \bar{y})^2) (\sum_{i=1}^n (x_i - \bar{x})^2)} \in [0, 1]$$

Logic

我们希望 generalize R^2 的定义, 使其适用于 y_i 为 scalar 但 x_i 为 vector 的情况

即考虑有 p 个 predictors 的情况, 我们希望能衡量 X matrix 的 usefulness

1.3 Lemma: Sum of Squares Decomposition

令:

- Regression 为 $Y \sim X$
- Design matrix 为 $X = (\vec{1} \quad X^*)$ (必须包括 intercept)
- Fitted value 为 $\hat{Y}$

则:

$$\underbrace{\sum_{i=1}^n (y_i - \bar{y})^2}_{\text{Total Sum of Squares (TotSS)}} = \underbrace{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}_{\text{Regression Sum of Squares (RegSS)}} + \underbrace{\sum_{i=1}^n (y_i - \hat{y}_i)^2}_{\substack{\text{Error Sum of Squares (ErrSS)} \\ \text{Residual Sum of Squares (RSS)}}$$

Remark: Regression Sum of Squares

对于 regression sum of squares, 若模型包含 intercept, 则其可被写作:

$$\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

因为此时

$$H\vec{1} = \vec{1} \Rightarrow \vec{1}^\top \hat{Y} = \vec{1}^\top H Y = \vec{1}^\top Y \Rightarrow \vec{1} \bar{y} = \frac{1}{n} \sum_{i=1}^n \hat{y}_i = \frac{1}{n} \sum_{i=1}^n y_i = \bar{y}$$

1.4 Definition: R^2 (Vector Version)

令:

- Regression 为 $Y \sim X$
- Design matrix 为 $X = (\vec{1} \quad X^*)$
- Fitted value 为 $\hat{Y}$

则 R^2 被定义为:

$$\begin{aligned} R^2 &= \frac{\text{RegSS}}{\text{TotSS}} = 1 - \frac{\text{ErrSS}}{\text{TotSS}} \\ &= \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \end{aligned}$$

△ Remark ▾

由于 Sum of Squares Decomposition, $R^2 \in [0, 1]$, 因此其可以被理解为 "proportion" of variation of Y that is explained by X

1.5 Another Formulation of R^2

考虑以下两个 regressions:

- Full Model: $Y \sim \vec{1}_n + X^*$
- Reduced Model: $Y \sim \vec{1}_n$

对于 reduced model, 有 $\hat{y} = \bar{y}$, 因此 $\text{ErrSS}_{\text{red}} = \text{TotSS}_{\text{red}} = \text{TotSS}_{\text{full}}$

因此 R^2 可以被定义为:

$$\begin{aligned} R^2 &= 1 - \frac{\text{ErrSS}(Y \sim \vec{1}_n + X^*)}{\text{ErrSS}(Y \sim \vec{1}_n)} \\ &= \frac{\text{ErrSS}(Y \sim \vec{1}_n) - \text{ErrSS}(Y \sim \vec{1}_n + X^*)}{\text{ErrSS}(Y \sim \vec{1}_n)} \end{aligned}$$

△ Remark: Intuition ▾

R^2 衡量了 full model 相比 mean-only model 的误差改进比例

△ Remark: Partial R^2 ▾

利用该 formulation, 可定义以下 partial R^2 :

$$R^2_{Y,X|W} = 1 - \frac{\text{ErrSS}(Y \sim \vec{1}_n + X + W)}{\text{ErrSS}(Y \sim \vec{1}_n + W)}$$

其可被理解为在已经控制 W 的情况下, 加入 X 后误差减少的比例

🔗 Logic ▾

默认情况下, F-statistics 用于检验除 intercept 外的所有 β_i 是否等于 0, 即 $H_0: \vec{\beta}^* = 0$, 由此可以和 R^2 的意义产生联系

1.6 Theorem: R^2 和 F-Statistics 的联系

令:

- Regression 为 $Y \sim X$
- Design matrix 为 $X = (\vec{1} \quad X^*)$
- F 为 F-statistics for $H_0 : \vec{\beta}^* = 0$

则:

$$F = \frac{n-p}{p-1} \cdot \frac{R^2}{1-R^2}$$

↪ **Proof** ∨

由于 $F_{\text{Wald}} = F_{\text{ANOVA}} = F$, 其中

$$F_{\text{ANOVA}} = \frac{(\text{ErrSS}_{\text{red}} - \text{ErrSS}_{\text{full}})/(p-1)}{\text{ErrSS}_{\text{full}}/(n-p)}$$

且由于

$$\begin{aligned} R^2 &= 1 - \frac{\text{ErrSS}_{\text{full}}}{\text{ErrSS}_{\text{red}}} \\ \Rightarrow \frac{R^2}{1-R^2} &= \frac{1 - \frac{\text{ErrSS}_{\text{full}}}{\text{ErrSS}_{\text{red}}}}{\frac{\text{ErrSS}_{\text{full}}}{\text{ErrSS}_{\text{red}}}} = \frac{\text{ErrSS}_{\text{red}} - \text{ErrSS}_{\text{full}}}{\text{ErrSS}_{\text{full}}} \end{aligned}$$

因此

$$F = \frac{n-p}{p-1} \cdot \frac{R^2}{1-R^2}$$